

Machine Learning

Introduction - Week 1

Objectives

- Personal Introductions
- Overview of Course/Syllabus
- What is Machine Learning?
 - Supervised vs Unsupervised
- Types of Data (reading assignment)

About me



You can call me **Dr. Banerjee!**

Where to find me?

Office:

Science Hall 250

Office Hours:

**Weds and Thurs 11:00 AM -
1:00 PM EST or by
appointment**



Tell me about Yourself

- Name?
- Major or Intended Major? (or not sure yet)
- What are your expectations from the course?
- Any hobbies? Pets?

TLDR;

- ▶ **No assigned textbook.** I will share notes.
- ▶ Class is a mix of **3 tests (including finals)** and a **project**. Details about the project will be shared soon.
- ▶ Assignments (mostly **coding exercises**) are assigned after a module is completed and will be based on what you learned. **These won't be graded.** However, they will help you with the exams.

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- ▶ A deep understanding of how ML works necessitates extensive background in Linear Algebra, Calculus and Statistics. Since we are not pre-supposing such math background, we will focus on the practical and computational aspects of the discipline.

On Plagiarism/Cheating

- ▶ **All work must be your own! This includes assignments/quizzes that you are assigned.**
- ▶ You may use online resources for reference, but sharing code with other students is considered plagiarism. This includes copying and pasting code from any source, including the internet.
- ▶ Any instance of plagiarism or cheating will result in a failing grade for the assignment.
- ▶ If plagiarism is detected in a single question, the entire assignment will receive a zero. Be aware that suspected cases of academic dishonesty will be reported to the Department Chair and the academic coordinator.

Hardware & Software

- ▶ You will need a **Windows, Mac, or Linux computer** where you can install **Python** along with the libraries, primarily NumPy, Pandas, Scipy, Sklearn, JupyterLab.
- ▶ Lessons will be presented in Jupyter Notebooks. Coding assignments should be submitted as Jupyter Notebooks with the following naming convention unless specified:
LastName_FirstName_Assignment# (ex. **Banerjee_Sreya_Assignment1**)
- ▶ **Having a stable coding environment is your responsibility!**

Github

- ▶ Most of my lessons will be shared in Github. You are encouraged to create your own github profile at <https://github.com/>
- ▶ Lessons will be presented in Jupyter Notebooks. Coding assignments should be submitted as Jupyter Notebooks with the following naming convention unless specified:
LastName_FirstName_Assignment# (ex. Banerjee_Sreya_Assignment1)
- ▶ **Having a stable coding environment is your responsibility!**

Acknowledgement

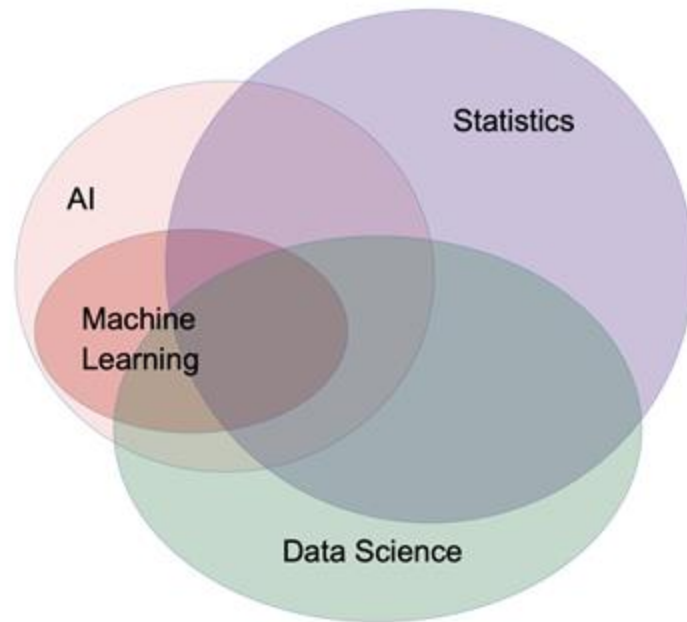
- ▶ Original Course design by:
 - ▶ **Dr. Chirakkal Easwaran**, SUNY New Paltz

What is Machine Learning?

Machine Learning is a term coined by an American IBM scientist, Arthur Samuel, in 1959.

At the time, it was just a new term for a sub-field of Artificial Intelligence.

Some people still consider Machine Learning a sub-field of AI, while others do not.



Let's look at some examples...

Given, some data on housing prices in rural Indiana, our **task** is to find the price of the a house with the following details:

Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3400	4	2.5	0	???

This is a **machine learning** problem.

Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3000	4	2	1	300K
2500	3	1.5	0	330K
1900	3	2	1	290k
2800	4	2	1	350K
3300	4	2	0	340K
.....				

Table: Inventory of houses sold in Rural Indiana

Let's look at some examples...

Note, The "learning" process here is constructing a model (essentially a formula) that can answer the Sales price question posed.

Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3000	4	2	1	300K
2500	3	1.5	0	330K
1900	3	2	1	290k
2800	4	2	1	350K
3300	4	2	0	340K
.....				

Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3400	4	2.5	0	???

Dependent variable/output variable/ target

Independent variable/input features

Let's look at some examples...

This is also known as Supervised Learning, where we have **available data** with well-defined **features and target variables** to predict new target variable based on some given features.

Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3000	4	2	1	300K
2500	3	1.5	0	330K
1900	3	2	1	290k
2800	4	2	1	350K
3300	4	2	0	340K
.....				

Available Data

Note that the model, we create is very specific to the type of problem and environment from which the data comes from.

I.e. The sales price of houses in NYC is going to be very different from rural Indiana. The same model would not give us accurate sales prices of houses in NYC.

Total Area	Number of bedrooms	Number of bathrooms	Has finished basement	Sales Price
3400	4	2.5	0	???

New Data

Other examples...

Data on a **certain group's decision to play or not**, based on various atmospheric factors, are recorded over time.

Using this data, can you **predict** the decision to play or not on a particular day?

This is also a **supervised machine learning problem** - A model is constructed from past behavior that is then used to predict an **outcome (Play? Yes or No)**.

In this example, the **output or target variable** is binary (takes one of two values),

The input variables are the atmospheric conditions.

Outlook	Temperature	Humidity	Windy	Play
overcast	hot	high	FALSE	yes
overcast	cool	normal	TRUE	yes
overcast	mild	high	TRUE	yes
overcast	hot	normal	FALSE	yes
rainy	mild	high	FALSE	yes
rainy	cool	normal	FALSE	yes
rainy	cool	normal	TRUE	no
rainy	mild	normal	FALSE	yes
rainy	mild	high	TRUE	no
sunny	hot	high	FALSE	no
sunny	hot	high	TRUE	no
sunny	mild	high	FALSE	no
sunny	cool	normal	FALSE	yes
sunny	mild	normal	TRUE	yes
rainy	cool	high	TRUE ???	

Other examples...

Task: Classifying email as Spam or not-Spam.

This is also an example of supervised learning. Can you think of the features here? What are those?

Clustering

The Problem: We've introduced a new product and have data on where people bought it.

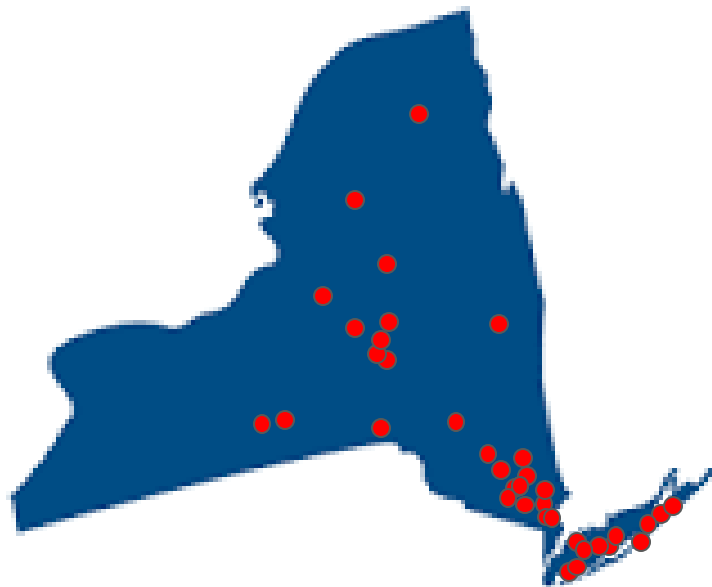
The Question: Are there specific neighborhoods or cities where our product is selling a lot more than in other places?

Why It Matters:

- If we find these **"hot spots"** we know where to spend our advertising money.
- We can also make other decisions, like sending more of the product to those areas or creating special promotions for local customers.

How We Do It:

- We can put the sales data on a map.
- Then, we can see if the sales "dots" are **clustered** or grouped together in certain areas.

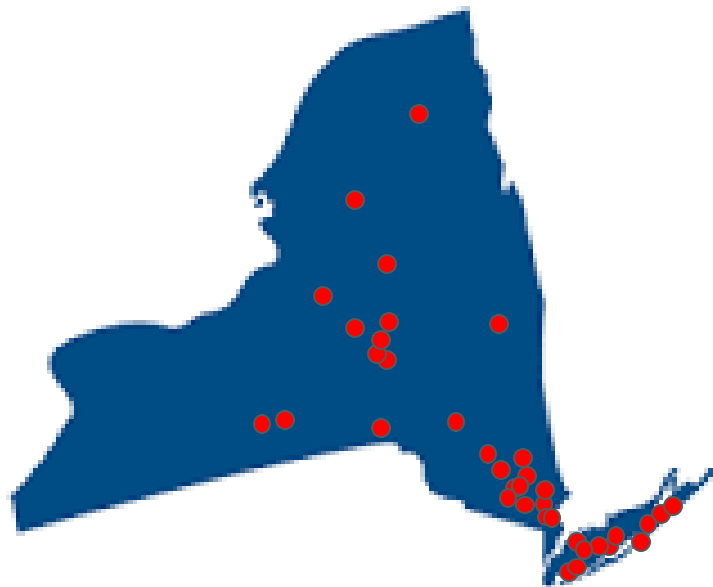


Clustering

Clustering, in general is an example of an **unsupervised learning algorithm**.

Here, we have well-defined features i.e. Transaction ID, date and time, number of products sold, location (latitude, longitude) but **no defined output variable**.

From the available data, we need to find the patterns of buying, which isn't **explicitly mentioned in the available data**. This is something **we need to derive**.



ANY QUESTIONS?